

Robust-DeepFake: Improving JPEG Compression Robustness in Deepfake Detection through Compression-Aware Training and Frequency-Aware Learning

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Abstract

Deepfake detection models have achieved remarkable performance on benchmark datasets; however, their effectiveness deteriorates considerably under real-world conditions involving JPEG compression and cross-dataset distribution shifts. In this work, we present a comprehensive robustness study of convolutional and transformer-based deepfake detectors using Xception65 and ViT-B/16 as representative baselines. The models are trained on FaceForensics++ and evaluated on both FaceForensics++ and Celeb-DF across multiple JPEG compression levels to systematically analyze the effects of image degradation and domain shift. While the baseline Xception65 achieves an AUC of 0.9547 on FaceForensics++, its performance drops to 0.7872 on Celeb-DF and further declines to 0.6318 under severe JPEG compression (Q10), highlighting the vulnerability of conventional CNN-based detectors.

To improve robustness against compression artifacts, we investigate two complementary strategies for CNN-based models: a DCT-aware Xception architecture that combines spatial and frequency-domain representations, and a Compression-Aware Training (CAT) framework that exposes the model to randomly compressed images during training. We also evaluate a consistency-regularized Vision Transformer to examine the effect of feature-level invariance under image compression. Experimental results show that the proposed CAT model substantially improves robustness under severe compression, increasing the AUC at JPEG quality 10 from 0.6318 to 0.7584, while the DCT-aware architecture consistently outperforms the baseline Xception across most compression levels. Furthermore, the consistency-regularized ViT improves cross-dataset performance on Celeb-DF, increasing the AUC from 0.7765 to 0.8000. Overall, our findings demonstrate that compression-aware learning and frequency-domain information significantly enhance CNN robustness, whereas transformer-based representations exhibit greater resilience to compression and domain shift.

1. Introduction

Deepfake generation has advanced rapidly with modern generative models, enabling highly realistic facial manipulations. While beneficial for creative applications, it raises serious concerns for digital security and forensic integrity, making robust detection increasingly important.

Despite strong results on benchmarks such as FaceForensics++ [7], existing detectors often fail in real-world settings due to post-processing operations like JPEG compression, resizing, and blurring. These operations remove subtle forensic cues such as blending artifacts and color inconsistencies that models rely on for detection.

A further challenge is poor cross-dataset generalization. Models trained on one dataset often degrade significantly when tested on others (e.g., FaceForensics++ [7] vs. Celeb-DF [4]) due to differences in manipulation methods, capture conditions, and compression pipelines.

To address these issues, we study robustness under compression and dataset shift using two baselines, Xception65 and ViT-B/16, and propose three lightweight improvements:

- **DCT-Aware Xception:** Incorporates frequency-domain (DCT) features with spatial RGB features for improved robustness.
- **Compression-Aware Training:** Uses random JPEG compression during training to improve invariance to quality degradation.
- **ViT-Consistency:** Enforces feature consistency between clean and compressed inputs for better generalization.

Robust-DeepFake: Analyzing Domain Shift and Enhancing JPEG Compression Robustness in Deepfake Detection

Project Repository: https://github.com/mubashar-ds/MSDS24031_Project_DLSpring2026

2. Related Work

Deepfake detection has become increasingly challenging as manipulation techniques grow more sophisticated. A growing body of work has revealed that existing detectors are

fragile when deployed in realistic scenarios involving image compression, noise, and other common processing operations [2] [5]. This section reviews relevant literature on robustness strategies for compressed deepfake detection.

2.1. Frequency-Domain Approaches

A significant direction in the literature addresses compression robustness through frequency-domain analysis. Qian et al. [6] proposed F3-Net, a framework that leverages frequency-aware clues through a two-stream architecture with Frequency-Aware Decomposition (FAD) and Local Frequency Statistics (LFS) branches. By mining subtle forgery patterns in the frequency domain, F3-Net demonstrated that utilizing DCT components can mitigate performance drops under compression. However, the method still faces challenges at extreme compression levels where quantization noise significantly corrupts frequency information.

Building on this paradigm, Tan et al. [8] introduced FreqNet (AAAI 2024), which addresses the generalization limitations of existing frequency-based detectors by forcing continuous focus on high-frequency information through phase and amplitude spectrum learning between FFT and iFFT. Their approach achieved state-of-the-art performance across 17 GANs with a 9.8% improvement over previous methods.

2.2. Consistency-Based Representation Learning

A complementary direction involves learning invariant representations through consistency-based training. Zha et al. [9] introduced real-centric consistency learning, which constrains feature extraction at both sample and feature levels to narrow intra-class discrepancies and widen inter-class gaps. Their method explicitly addresses robustness challenges posed by internet compression by learning invariant representations for both natural and forged faces.

2.3. Vision Transformers for Robust Detection

Recent work has also explored Vision Transformer (ViT) architectures for deepfake detection, particularly for their inherent robustness to compression artifacts. Studies show that while frequency-aided CNNs excel on clean images, they exhibit substantial degradation under compression, whereas ViTs maintain robustness through semantic pattern recognition [1]. Khan and Jion [3] demonstrated that ViT-based models degrade minimally (92.86% to 88.36%) under JPEG compression compared to CNN-based approaches (93.33% to 61.49%), indicating that transformer architectures offer inherent robustness benefits for compressed media detection.

2.4. Research Gap and Motivation

Despite these advances, existing approaches face persistent limitations. Frequency-aware methods demonstrate strong

performance on moderate compression but struggle at extreme levels where quantization noise significantly degrades frequency information. Furthermore, while consistency learning has shown promise for compression-invariant representations, few studies systematically compare its effectiveness against simpler augmentation strategies within the same architectural framework.

Our work addresses this gap by systematically comparing three XceptionNet-based approaches: a standard baseline, a JPEG-augmented model, and a consistency-trained model. Additionally, we explore a ViT-based consistency approach to leverage the inherent robustness of transformer architectures. This comparative analysis contributes to understanding the relative effectiveness of different robustness strategies for real-world deployment scenarios.

3. Dataset and Preprocessing

We conduct experiments using two widely adopted deepfake benchmarks: FaceForensics++ (FF++) [7] for training and in-distribution evaluation, and Celeb-DF (v2) [4] for cross-dataset generalization testing.

3.1. Raw Data Extraction and Class Balancing

To construct a compact yet diverse dataset (approximately 10–15 GB), we adopt a frame-based sampling strategy. For FaceForensics++, we select 400 videos in total (300 for training, 50 for validation, and 50 for testing). To enforce strict class balance, we extract 25 frames per real video. For manipulated videos, which span five editing techniques (Deepfakes, Face2Face, FaceSwap, FaceShifter, and NeuralTextures), we extract 5 frames per method per video.

Celeb-DF (v2) is reserved exclusively for cross-dataset evaluation to assess robustness under distribution shift.

Frame extraction is performed using segment-based random sampling. Each video is divided into N equal temporal segments, and a single frame is randomly selected from each segment. This strategy improves diversity in facial pose, expression, and motion dynamics.

3.2. Face Detection, Cropping, and Normalization

Raw video frames include background regions that may introduce dataset-specific shortcuts. To mitigate this, we apply a structured preprocessing pipeline:

1. **Face Detection:** We use InsightFace’s RetinaFace detector (ResNet50-based model) to localize facial regions on GPU.
2. **Bounding Box Expansion:** To preserve manipulation cues near boundary regions such as the jawline, ears, and hairline, the detected bounding box is expanded by a margin of $0.3 \times$ the original dimension.
3. **Resizing:** Cropped faces are resized using area interpolation (`cv2.INTER_AREA`). Inputs are resized to

299×299 for Xception-based models and 224×224 for Vision Transformer-based models.

4. **Storage:** Processed images are saved using JPEG format with a quality factor of 95.

After preprocessing, the final dataset consists of 14,369 training frames, 2,380 validation frames, and 2,388 test frames from FF++, along with 5,990 test frames from Celeb-DF.

4. Methods

We investigate the robustness of deepfake detection models under compression and distribution shift using two representative architectures, Xception65 and ViT-B/16. Building upon these baselines, we propose four complementary strategies that improve robustness through frequency-domain modeling, compression-aware learning, and representation regularization.

4.1. Baseline Models

We consider two widely adopted deepfake detection architectures. Xception65 serves as the CNN baseline and extracts discriminative spatial features using depthwise separable convolutions. ViT-B/16 serves as the transformer baseline, partitioning images into fixed-size patches and modeling global dependencies through self-attention mechanisms.

4.2. DCT-Aware Xception

CNN-based detectors often rely on high-frequency artifacts that are vulnerable to JPEG compression. To address this limitation, we augment Xception65 with a frequency-aware branch that operates on Discrete Cosine Transform (DCT) representations of the input image. This auxiliary branch captures spectral forgery cues that may be less evident in the spatial domain. Features extracted from both RGB and DCT representations are fused prior to classification, enabling the model to leverage complementary spatial and frequency information.

Figure 1 presents the architecture of the proposed DCT-Aware Xception model.

4.3. Compression-Aware Training (CAT)

To improve robustness under varying image quality conditions, we introduce Compression-Aware Training (CAT). During training, images are randomly subjected to JPEG compression using different quality factors. This exposes the model to a broad range of degradation levels and discourages over-reliance on fragile high-frequency artifacts present only in clean samples. As a result, the learned representations become more stable across real-world transmission and storage conditions.

Figure 2 illustrates the proposed Compression-Aware Training (CAT) pipeline.

4.4. Synergy Between Frequency Fusion and Compression-Aware Training

Frequency-domain modeling and compression-aware learning address complementary aspects of robustness. While DCT representations improve detection performance by exposing spectral forgery artifacts, frequency information alone does not guarantee resilience to severe compression. When trained exclusively on clean samples, the frequency branch may learn to exploit highly discriminative but compression-sensitive coefficients that disappear after aggressive JPEG quantization.

Compression-Aware Training mitigates this limitation by exposing the network to degraded samples during optimization. Consequently, the model learns to focus on stable spectral patterns rather than transient high-frequency noise. The combination of DCT fusion and CAT therefore creates a synergistic effect: the DCT branch provides informative frequency-domain features, while CAT encourages the network to learn compression-invariant representations within this feature space. This combination enables strong performance on clean data while substantially improving robustness under severe compression.

Figure 3 illustrates the proposed DCT-Compression-Aware Xception architecture.

4.5. Consistency-Regularized Vision Transformer

While data augmentation improves robustness, it does not explicitly enforce invariance in the learned feature space. To address this, we introduce a consistency-based training strategy for ViT-B/16. The model processes both an original image and its compressed counterpart and is encouraged to produce similar feature representations for both inputs. This regularization promotes compression-invariant embeddings and improves generalization to unseen datasets and degradation levels.

5. Experiments

5.1. Experimental Setup

All models are implemented in PyTorch using the `timm` library and trained with mixed-precision on NVIDIA T4/A100 GPUs. We use the AdamW optimizer with weight decay 10^{-4} . The learning rate is set to 5×10^{-5} for ViT and 10^{-4} for Xception, scheduled using ReduceLROnPlateau (factor 0.5, patience 2). The batch size is 32, and training is limited to 20 epochs with early stopping (patience = 5). Training time is approximately 15 minutes for Xception and 91.7 minutes for ViT-Consistency.

5.2. Robustness under JPEG Compression (FF++ Evaluation)

We evaluate all models on FaceForensics++ under varying JPEG compression levels: Clean, Q90, Q70, Q50, Q30, and

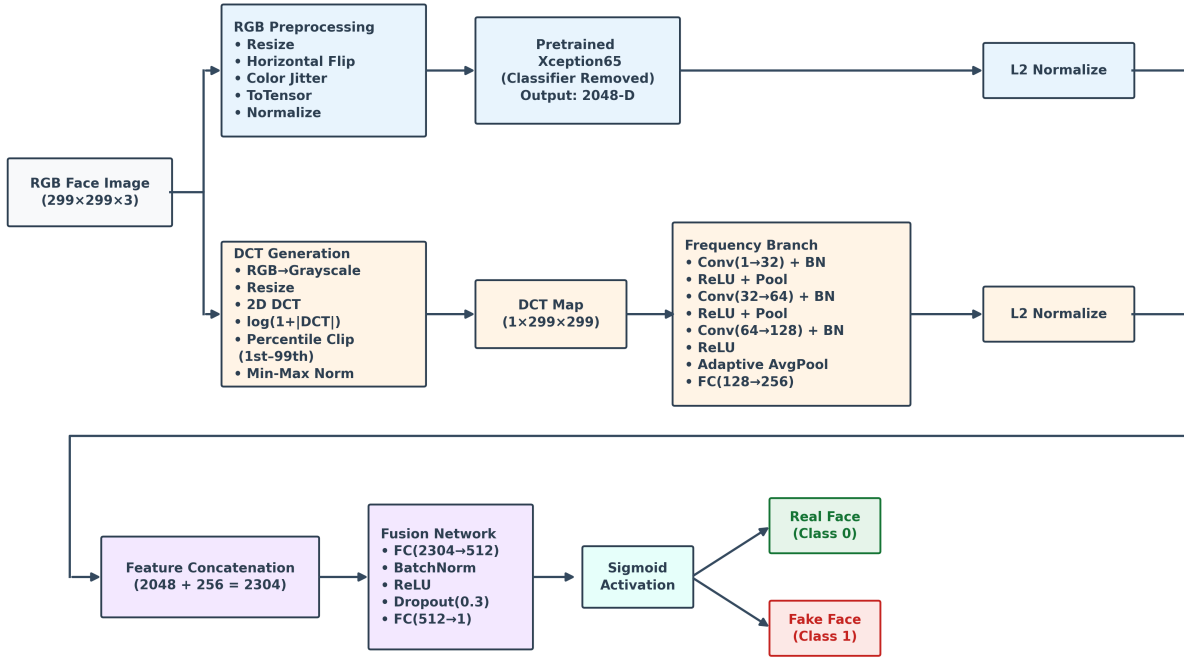


Figure 1. Overview of the proposed DCT-Aware Xception architecture. The input image is processed through two complementary branches: an RGB branch that extracts spatial features using a pretrained Xception65 backbone, and a frequency branch that extracts DCT log-magnitude features using a lightweight CNN. The normalized feature representations from both branches are concatenated and fused through fully connected layers to produce the final real/fake prediction, enabling the model to leverage both spatial and frequency-domain information for improved robustness to JPEG compression artifacts.

Q10. Table 1 summarizes AUC and F1 scores.

Findings. CNN-based models suffer severe degradation under compression, with the baseline Xception collapsing at Q10 ($F1 = 0.0122$). Frequency modeling (DCT) improves clean performance but does not fully resolve robustness issues. Compression-aware training significantly stabilizes performance under extreme degradation. The proposed DCT-CAT-Xception achieves the best CNN robustness at Q10 ($F1 = 0.6683$). In contrast, ViT exhibits strong inherent robustness, degrading only marginally under compression due to its global attention mechanism.

5.3. Cross-Dataset Generalization (Celeb-DF)

We further evaluate generalization on Celeb-DF under the same compression settings (Table 2).

Findings. All CNN baselines show significant performance drops under dataset shift, confirming poor generalization. Compression-aware training improves stability across all conditions. The ViT-Consistency model achieves the best overall transferability, improving both clean and

compressed performance, particularly under extreme degradation.

5.4. Ablation Study and Discussion

The ablation study highlights the contribution of each component.

- **DCT vs. CAT:** Frequency-domain features improve representation quality but do not ensure robustness alone. Compression-aware training is essential for invariance to image degradation.
- **Combined Effect:** The DCT-CAT-Xception achieves the strongest CNN performance under extreme compression, though with a slight trade-off on clean accuracy.
- **CNN vs. Transformer Robustness:** CNNs rely on local texture artifacts, which are highly sensitive to JPEG compression. In contrast, Vision Transformers leverage global context through self-attention, making them inherently more stable under degradation.

6. Conclusion

This work investigates the robustness of deepfake detection models under realistic degradation settings, including JPEG

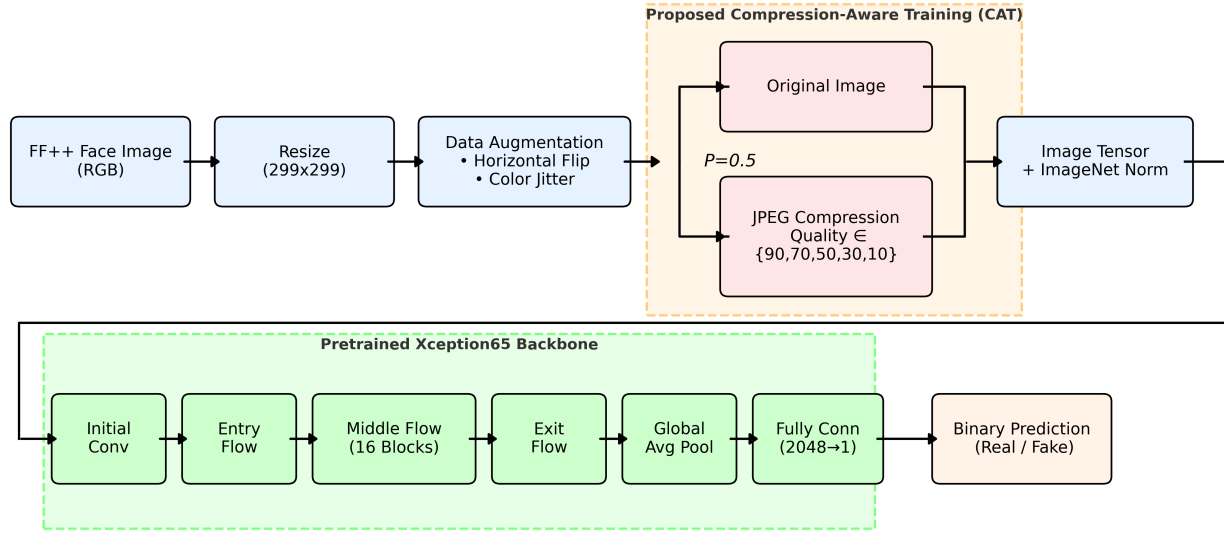


Figure 2. Overview of the proposed Compression-Aware Training (CAT) pipeline. During training, each input image is randomly compressed using one of several JPEG quality levels before being passed to the Xception65 network. Exposing the model to diverse compression artifacts encourages the learning of compression-invariant features and improves robustness to real-world JPEG degradation.

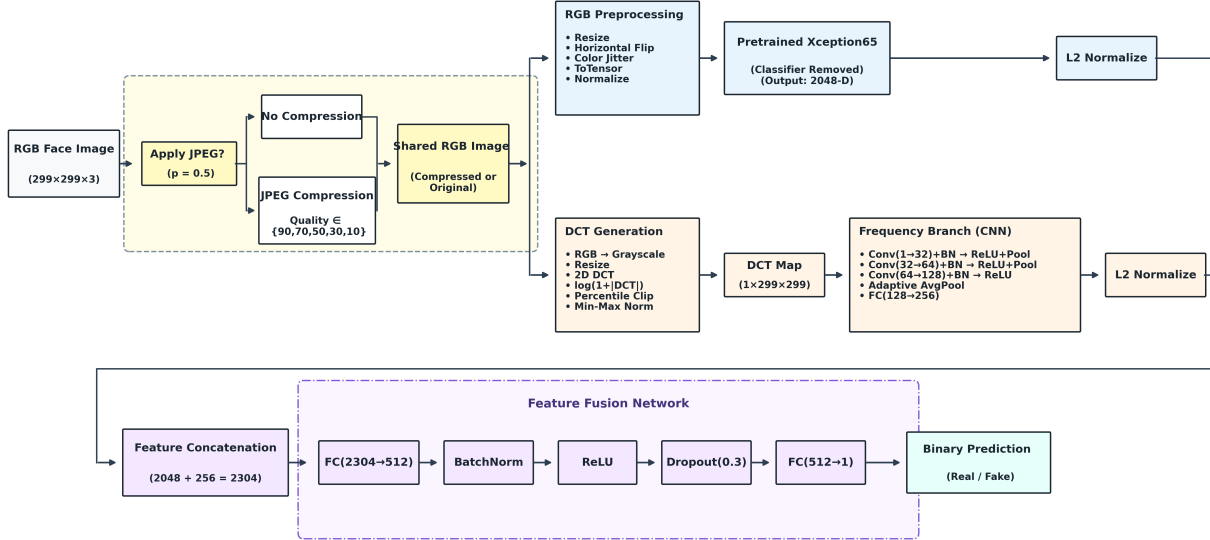


Figure 3. Overview of the proposed DCT-Compression-Aware Xception architecture, which combines Compression-Aware Training (CAT) with frequency-aware feature learning. During training, each input image is randomly compressed using one of several JPEG quality levels before being processed. The compressed image is then fed into two complementary branches: an RGB branch based on a pretrained Xception65 backbone for spatial feature extraction and a frequency branch that extracts DCT log-magnitude features using a lightweight CNN. The normalized feature representations are concatenated and fused through fully connected layers to produce the final real/fake prediction. By integrating compression-aware augmentation with joint spatial-frequency feature learning, the model is designed to improve robustness against JPEG compression artifacts while preserving strong detection performance.

Model	Clean	Q90	Q70	Q50	Q30	Q10
Xception Baseline	0.9547 / 0.8963	0.9467 / 0.8780	0.9140 / 0.7992	0.8785 / 0.6921	0.8211 / 0.4107	0.6318 / 0.0122
DCT-Xception	0.9657 / 0.9049	0.9591 / 0.8933	0.9367 / 0.8314	0.9004 / 0.7292	0.8335 / 0.4752	0.6463 / 0.0589
CAT-Xception	0.9452 / 0.8708	0.9289 / 0.8598	0.9436 / 0.8511	0.9245 / 0.8120	0.8942 / 0.7242	0.7838 / 0.5237
DCT-CAT-Xception	0.9388 / 0.8720	0.9340 / 0.8644	0.9309 / 0.8544	0.9149 / 0.8389	0.8878 / 0.8085	0.7912 / 0.6683
ViT Baseline	0.8839 / 0.7851	0.8843 / 0.7863	0.8808 / 0.7801	0.8734 / 0.7780	0.8705 / 0.7727	0.8135 / 0.7375
ViT Consistency	0.8889 / 0.7979	0.8880 / 0.7969	0.8840 / 0.7949	0.8807 / 0.7864	0.8740 / 0.7773	0.8192 / 0.7185

Table 1. Performance on FF++ under JPEG compression (AUC / F1).

Model	Clean	Q90	Q70	Q50	Q30	Q10
Xception Baseline	0.7872 / 0.7362	0.7840 / 0.7428	0.7813 / 0.7318	0.7482 / 0.6446	0.6975 / 0.3474	0.5593 / 0.0040
DCT-Xception	0.7885 / 0.7462	0.7881 / 0.7380	0.7833 / 0.7345	0.7517 / 0.6450	0.6976 / 0.3093	0.5534 / 0.0305
CAT-Xception	0.8039 / 0.7458	0.8007 / 0.7435	0.7973 / 0.7531	0.7853 / 0.7508	0.7606 / 0.7127	0.6838 / 0.4789
DCT-CAT-Xception	0.7921 / 0.7258	0.7810 / 0.7226	0.7852 / 0.7325	0.7741 / 0.7311	0.7562 / 0.7226	0.6888 / 0.6807
ViT Baseline	0.7765 / 0.7134	0.7766 / 0.7135	0.7754 / 0.7146	0.7743 / 0.7170	0.7711 / 0.7169	0.6904 / 0.6904
ViT Consistency	0.8000 / 0.7320	0.8003 / 0.7321	0.7990 / 0.7339	0.7964 / 0.7363	0.7935 / 0.7354	0.7174 / 0.7023

Table 2. Cross-dataset performance on Celeb-DF (AUC / F1).

compression and cross-dataset distribution shifts. Our analysis reveals that standard CNN-based detectors are highly sensitive to compression artifacts, with severe performance collapse under extreme degradation (Q10). This highlights a critical gap between benchmark performance and real-world deployment reliability.

To address this limitation, we propose three lightweight and complementary strategies: a dual-branch DCT fusion mechanism to incorporate frequency-aware representations, a compression-aware training (CAT) strategy to improve invariance to image quality variations, and a consistency-based regularization framework to stabilize feature embeddings under input perturbations. Collectively, these approaches significantly improve robustness across both in-distribution and out-of-distribution settings. In particular, the ViT-Consistency model demonstrates strong cross-dataset generalization on Celeb-DF while maintaining competitive clean performance.

Future Work. Several directions can further extend this study:

- **Temporal Consistency:** Extending consistency constraints from static images to video sequences to enforce temporal coherence across frames.
- **Curriculum Compression Learning:** Gradually increasing compression severity during training to improve stability under progressive degradation.
- **Broader Corruptions:** Evaluating robustness under additional real-world distortions such as motion blur, resolution downsampling, and video compression artifacts (e.g., H.264 and H.265 codecs).

7. Team Contributions

The project was carried out collaboratively, with all team members participating in planning, discussions, and reviewing experimental results. Individual members took primary responsibility for different implementation and evaluation tasks as follows:

Marriam Rasib (MSDS24027): Contributed to the implementation and evaluation of the Vision Transformer models, assisted with the compression-aware training experiments, and participated in analyzing the experimental results and preparing project documentation.

Mubashar Hussain (MSDS24031): Led the implementation and integration of the complete deepfake detection pipeline. Responsibilities included dataset preparation and preprocessing, implementation of the Xception65 baseline, development of the proposed DCT-aware, Compression-Aware (CAT), and combined CAT+DCT Xception architectures, training and evaluation of all Xception-based models, JPEG compression robustness experiments, cross-dataset evaluation on FF++ and Celeb-DF, comparative performance analysis, visualization of results, and preparation of the architecture diagrams and experimental workflow.

Saeed Anwer (MSDS24056): Implemented the Vision Transformer consistency regularization approach, contributed to preprocessing and dataset preparation, assisted in conducting experiments, and evaluated the robustness of the transformer-based models under different JPEG compression levels.

Syed Ali Raza Shah (MSDS24062): Assisted with implementing the baseline evaluation pipeline, generating performance metrics and visualizations, organizing experimental outputs, and contributing to result verification, report preparation, and group discussions.

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